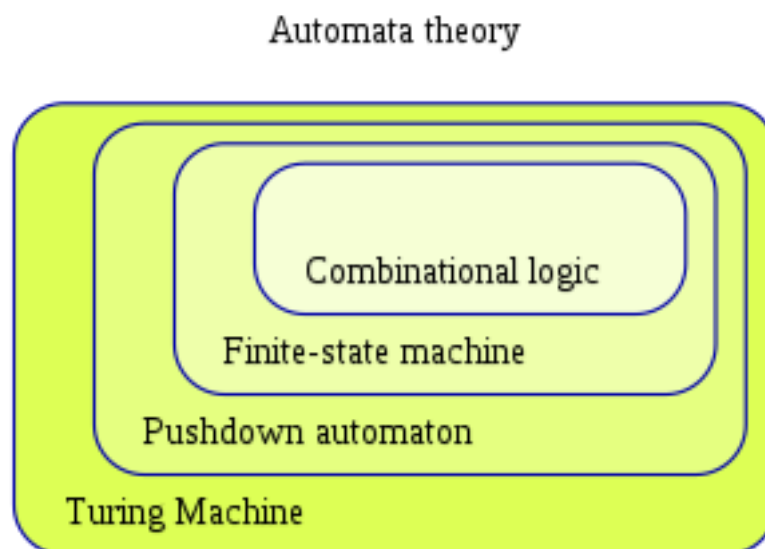


Theory of Automata

Theory of Automata

What is Theory of Automata ?

- Automata theory is the study of **abstract machines** and **automata**, as well as the **computational problems** that can be solved using them.
- It is a theory in **theoretical computer science**. The word automata (the plural of automaton) come from the Greek word αὐτόματα, which means "self-making".
- An automaton (Automata in plural) is an abstract self-propelled computing device which follows a **predetermined sequence of operations automatically**. An automaton with a finite number of states is



called a Finite Automaton (FA) or Finite State Machine (FSM).

- **Automata Theory** is a branch of **computer science** that deals with designing abstract **self-propelled computing devices** that follow a **predetermined sequence of operations automatically**. An automaton with a finite number of states is called a Finite Automaton.

- **Theory of Automata** General Introduction: It is a theory of Mathematical subject. In general, this subject focusses on the theoretical aspects of Computer Science.
- **Theory of Automata Applications** General Introduction.
 - Theory of Computation.
 - Compiler Constructions.
 - Defining Computer Languages.
- In this course, we will discuss theory of automata and defining computer languages.
- Compiler constructions is out of scope of this subject.
- **Types of Languages**
 - Formal Languages. For example, Computer languages, C, C++, Java.
 - Informal Languages. English, Urdu etc.
- **Elements of a Language**
 - Alphabets , letters.
 - Strings.
 - Words.
 - Palindromes. madam
- **Sets**
 - Union of Sets.
 - Intersection of Sets.
 - Set Difference.
 - Complement.
- **Additional Terminology**
 - **Disjoint Sets.**
 $A = \{a, b, c, d\}$, and
 $B = \{w, x, y, z\}$, if $A \cap B = \text{Empty Set}$, A and B called disjoint sets.
 - **Cardinality.**
 $|A|$, number of elements in a set, or $|w|$, number of alphabets in a string.
 - $|A| = |\{a, b, c, d\}| = 4$

- $w = \text{university}$, $|w| = 10$
- **Powerset.**
If a set having 5 elements, its powerset will equal to 2^5 , means 32 sets.
 $A = \{1, 2, 3\}$, find $P(A)$.
Number of elements = 3,
Powerset $P(A) = 2^3 = 8$ sets.
 $P(A) = \{\text{empty set}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{3, 1\}, \{1, 2, 3\}\}$
- If $B = \{a, b, c, d\}$, find $P(B)$
- Number of elements = 4,
Powerset $P(A) = 2^4 = 16$ sets.
- $P(B) = \{\{\}, \{a\}, \{b\}, \{c\}, \{d\}, \{a, b\}, \{a, c\}, \{a, d\}, \{b, c\}, \{b, d\}, \{c, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, \{b, c, d\}, \{a, b, c, d\}\}$
- **Cartesian Product.**
 $A = \{1, 2\}$, $B = \{x, y, z\}$,
 $A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$

 $A = \{1, 2\}$, $B = \{x, y, z\}$, $C = \{3, 4\}$,
 $A \times B \times C = \{(1, x, 3), (1, x, 4), (1, y, 3), \dots\}$

• **Strings and Languages**

- **Alphabets**
 - Roman, $\{a, b, c, d, \dots, z\}$
 - Binary, $\{0, 1\}$
- **String:** Finite sequence of symbols from the alphabets.
- **Length of String:** $|10011| = 5$
- **Empty String:** String having length equal to 0, is called Lambda, λ .
- **Reverse String:** $w = abc$, reverse of $w = cba$.
- **Sub String:** "deck" is a sub-string of "abcdeckabb". "Sun" is sub-string of "Sunday"

- **Concatenation:** If x and y are two strings, concatenation of x and y is written as xy , which is equal to the string obtained by appending y at the end of x .
- $x = 0011$
- $y = aabbaa$
- $xy = 0011aabbaa$

Defining Alphabets:

- While defining an alphabet, an **alphabet** may contain **letters** consisting of group of **symbols**.
- **For example:** if alphabets $\Sigma = \{B, aB, bab, d\}$, then
 - B, aB are letters and
 - a, B, d are symbols.
- $\Sigma = \{B, Ba, bab, d\}$
 - $s = BababB$, is a string, check left to right:
 - $(Ba)(bab)(B)$, the string belongs to alphabets.
 - $s = BababB$, is a string, check **left to right**:
 - $(B)(abab)(B)$, $abab$ is not a proper letter. Thus, problem occurred, means, alphabets are not defined properly.
- **Remember:** while defining an alphabet of letters consisting of more than one symbols, no letter should be started with the letter of the same alphabet. Means, letter should not be prefix of another. However, a letter may be ended in the letter of same alphabet. Means, one letter may be the suffix of another.
 - Example: $\Sigma = \{B, Ba, bab, d\}$ is **invalid** alphabet.
 - However, $\Sigma = \{B, aB, bab, d\}$ is a **valid** alphabet.
- **Length of Strings:** The length of string s is denoted by $|s|$, is the number of letters in a string.
 - **Example1:** if $\Sigma = \{a, b\}$ and $s = ababa$, then $|s| = 5$.
 - **Example2:** if $\Sigma = \{B, aB, bab, d\}$, $s = BaBbabBd$, then $|s| = 5$.
[check it from left to right $(B)(aB)(bab)(B)(d)$]
 - **Example3:** if $\Sigma = \{B, aB, bab, d\}$, $s = BaBbabBd$, then $|s| = 8$.

- **Reverse of Strings:** The reverse of a string s denoted by $\text{Rev}(s)$ or s^r , is defined by writing the letters of s in reverse order.
 - **Example1:** if $s = abc$ is a string over alphabets $\Sigma = \{a, b, c\}$, then $\text{Rev}(s) = s^r = cba$.
 - **Example2:** if $s = BaBbabBd$ is a string over alphabets $\Sigma = \{B, aB, bab, d\}$, then $\text{Rev}(s) = s^r = dBbabBaB$ is **incorrect**, however, $s^r = dBbabaBB$ is **correct**. [(d) (B) (bab) (aB) (B)]

Kleen Star Closure: (it is an operator).

- For a given Σ , the Kleen Star Closure of the alphabets Σ is denoted by Σ^* .
- It is the collection of **all** the strings over Σ , including **lambda** (means; all possible concatenation of all strings).
- Kleen star closure can also be defined over any set of strings.
- **Example1:** if $\Sigma = \{a\}$, then $\Sigma^* = \{\lambda, a, aa, aaa, aaaa, aaaaa, \dots\}$ infinite language.
- **Example2:** if $\Sigma = \{0,1\}$, then $\Sigma^* = \{\lambda, 0, 1, 00, 01, 10, 11, \dots\}$
- **Example3:** if $\Sigma = \{aB, c\}$, then $\Sigma^* = \{\lambda, aB, c, aBaaB, aBc, caaB, cc, \dots\}$ infinite strings.
- Note: Languages generated by Kleen star closure of set of strings, are **infinite** languages. By infinite language, it is supposed that the language contains infinite strings, each of finite length.

Mix Examples:

Example: If $\Sigma = \{aB, c\}$, $\Sigma^7 = ?$, $\Sigma^7 = \{cccccc, aBcccccc, \dots\}$

Example: If $\Sigma = \{aB, c\}$, $\Sigma^9 = ?$

Example: $(aB)^* = ?$ $L = \{\lambda, aB, aBaaB, aBaaBaaB, aBaaBaaBaaB, \dots\}$

Example: $1^* = ?$, $L = \{\lambda, 1, 11, 111, 1111, 11111, \dots\}$

Example: $1^{10} = ?$, $L = \{1111111111\}$

Example: $1^0 = ?$ $L = \{\lambda\}$, is it empty string or not? It is not empty language.

PLUS Operation (+)

- Plus Operation is same as Kleene Star Closure except that it does not generate (null string), automatically.
- $a^+ = (a)(a^*)$

- **Example**
If $\Sigma = \{0,1\}$
Then $\Sigma^+ = \{0, 1, 00, 01, 10, 11, \dots\}$
- **Example,**
If $\Sigma = \{aab, c\}$
Then $\Sigma^+ = \{aab, c, aabaab, aabc, caab, cc, \dots\}$

Remark

It is to be noted that Kleene Star can also be operated on any string i.e. a^* can be considered to be all possible strings defined over $\{a\}$, which shows that a^* generates $\lambda, a, aa, aaa, \dots$

It may also be noted that a^+ can be considered to be all possible non empty strings defined over $\{a\}$, which shows that a^+ generates $a, aa, aaa, aaaa, \dots$

Definition of Language:

- Any **set of Strings** over an **alphabet** is called **Language**.
- Defining Languages: There are **four** ways:
 - Descriptive Definition.
 - Recursive Definition.
 - By using Regular Expression (RE).
 - By using Finite Automata (FA).

Descriptive Definition

The language is defined by describing the conditions imposed on its string.

Example:

The language L of strings of odd length, defined over $\Sigma = \{a\}$, can be written as
 $L = \{a, aaa, aaaaa, aaaaaaa, \dots\}$

Example:

The language L of strings of even length, defined over $\Sigma = \{a\}$, can be written as

$L = \{\lambda, aa, aaaa, aaaaaa, aaaaaaaa, \dots\}$

Example:

The language L of strings that does not start with a, defined over $\Sigma = \{a, b, c\}$, can be written as

$L = \{b, c, ba, bb, bc, ca, cb, cc, \dots\}$

Example:

The language L of strings of length 2, defined over $\Sigma = \{0, 1, 2\}$, can be written as

$L = \{00, 01, 02, 10, 11, 12, 20, 21, 22, \dots\}$ infinite or finite?

(Number of letters in alphabets) $^{(\text{Length})} = 3^2 = 9$; Means total 9 strings are possible in this language.

Example:

The language L of strings ending with 0, defined over $\Sigma = \{0, 1\}$, can be written as

$L = \{0, 00, 10, 000, 010, 100, 110, \dots\}$ infinite or finite?

Example: EQUAL

The language L (EQUAL language) of strings with number of a's equal to number of b's, defined over $\Sigma = \{a, b\}$, can be written as

$L = \{\lambda, ab, aabb, abab, baba, abba, baab, aaabbb, \dots\}$ infinite or finite?

Example: EVEN-EVEN Language

The language, EVEN-EVEN, of strings with even number of a's and even number of b's, defined over $\Sigma = \{a, b\}$, can be written as

$L = \{\lambda, aa, bb, aaaa, aabb, abab, abba, baab, baba, bbaa, bbbb, \dots\}$

Example: INTEGER

The language INTEGER, of strings defined over $\Sigma = \{-, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ can be written as

$L = \{\dots, -2, -1, 0, 1, 2, 3, 4, \dots\}$

[here – is a symbol, we can use it in letters, while defining alphabets]

Example:

The language **EVEN**, of strings defined over $\Sigma = \{-, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ can be written as

$L = \{\dots, -4, -2, 0, 2, 4, 6, \dots\}$

[Means: set of even integers]

Example:

The language $\{a^n b^n\}$, of strings defined over $\Sigma = \{a, b\}$,

As $\{a^n b^n : n = 1, 2, 3, 4, 5, \dots\}$, can be written as

$L = \{ab, aabb, aaabbb, aaaabbbb, \dots\}$

Example:

The language $\{a^n b^n a^n\}$, of strings defined over $\Sigma = \{a, b\}$, as $\{a^n b^n a^n : n = 1, 2, 3, 4, \dots\}$, can be written as

$L = \{aba, aabbaa, aaabbbbaaa, aaaabbbbbaaaa, \dots\}$

Example:

The language **factorial**, of strings defined over $\Sigma = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$, can be written as

$L = \{1!, 2!, 3!, 4!, 5!, \dots\}$

or

$L = \{1, 2, 6, 24, 120, \dots\}$

[factorial is possible only for positive integers]

Example:

The language **FACTORIAL**, of strings, defined over $\Sigma = \{a\}$,

As $\{a^{n!} : n = 1, 2, 3, 4, \dots\}$, can be written as

$L = \{a, aa, aaaaaa, aaaaaaaaaaaaaaaaaaaaaaa, \dots\}$

[It is to be noted that, the language **FACTORIAL**, can be defined over any **single letter** alphabet].

Example:

The language **DOUBLE FACTORIAL**, of strings defined over alphabets $\Sigma = \{a, b\}$, as

$\{a^{n!} b^{n!} : n = 1, 2, 3, 4, \dots\}$, can be written as

$L = \{aa, aabb, aaaaaabbbbbbb, \dots\}$

[Hint: for, $n=1$ and $n!=1$, $n=2$ and $n!=2$, $n=3$ and $n!=6$, so-on]

Example:

The language **SQUARE**, of strings defined over alphabets $\Sigma = \{a\}$,

As $\{a^{n^2} : n = 1, 2, 3, 4, 5, \dots\}$, can be written as,

$L = \{a, aaaa, aaaaaaaaa, \dots\}$

[Hint: $1a$, $4a$'s, $9a$'s, $16a$'s,.....]

Example:

The language **DOUBLE-SQUARE** of strings, defined over $\Sigma = \{a, b\}$,

As $\{a^{n^2} b^{n^2} : n = 1, 2, 3, 4, \dots\}$, can be written as

$L = \{ab, aaaabbbb, aaaaaaaaaabbbbbbbbbb, \dots\}$

[Hint: $1a1b$, $4a$'s $4b$'s, $9a$'s $9b$'s,]

Example:

The language **PRIME**, of strings, defined over $\Sigma = \{a\}$,

As $\{a^P : P = \text{is a prime number}\}$, can be written as

$L = \{aa, aaa, aaaaa, aaaaaaa, aaaaaaaaa, aaaaaaaaaaa, aaaaaaaaaaaa, \dots\}$

[Hint: Prime numbers 2, 3, 5, 7, 11, 13,..... **As 1 is not a prime number**]

PALINDROME

A language consisting of λ and the strings defined over $\Sigma = \{a, b\}$, such that $\text{Rev}(s) = s$.

aba, aba, aabbaa, aabbaa, aabaa, bbabb

Example:

For $\Sigma = \{a, b\}$,

PALINDROME = $\{\lambda, a, b, aa, bb, aaa, aba, bab, bbb, \dots\}$, infinite or finite?

Question: How to calculate number of strings of a specified length, defined over a given Σ ?

Answer: The number of strings of length " m " defined over alphabets of " n " letters is n^m .

For example: No. of strings of length 3, defined over $\Sigma = \{a, b\}$, as number of letters in alphabets is equal to 2,

thus, No. of strings having length 3 = 2^3 .

$L = \{aaa, aab, aba, baa, abb, bab, bba, bbb\}$, here $m = 8$

Total 8 strings.

If $\Sigma = \{a, b, c, \dots, z\}$

String length = 2.

Total number of strings = 26^2

Palindromes

The language consisting of Λ and the strings s defined over Σ such that $\text{Rev}(s)=s$.

It is to be denoted that the words of PALINDROME are called palindromes.

Example For $\Sigma=\{a,b\}$, $\text{PALINDROME}=\{\Lambda, a, b, aa, bb, aaa, aba, bab, bbb, \dots\}$

Length of Palindromes:

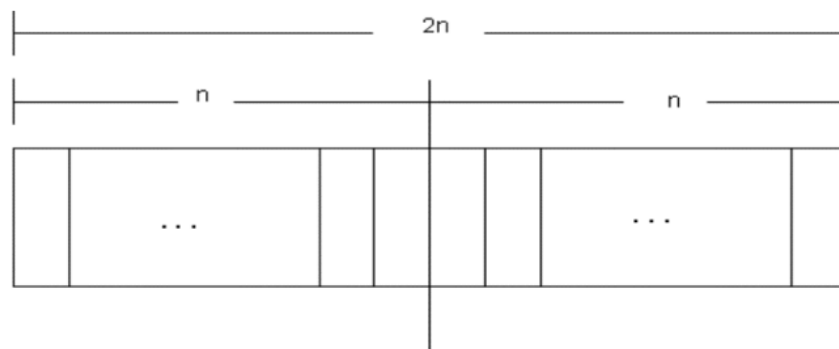
- **Even** length Palindromes. aabbaa
- **Odd** length Palindromes. aabaa

Even Length Case:

Examples The language of strings of length 2, defined over $\Sigma=\{a,b\}$ is $L=\{aa, ab, ba, bb\}$ i.e. number of strings = 2^2

The language of strings of length 3, defined over $\Sigma=\{a,b\}$ is $L=\{aaa, aab, aba, baa, abb, bab, bba, bbb\}$ i.e. number of strings = 2^3

To calculate the number of palindromes of length $(2n)$, consider the following diagram,



There are as many Palindromes of length “ $2n$ ” as there are strings of length n .

No. of palindromes of length $2n$ = No. of strings of length n

No. of palindromes of length 2 = No. of strings of length 1

No. of palindromes of length 4 = No. of strings of length 2

No. of palindromes of length 6 = No. of strings of length 3

Means: If we have 8 strings of length 3, then we will have 8 palindromes of length 6. (for $n = 3$).

How to prove above?

No. of strings of **length 3**, defined over $\Sigma = \{ a, b \}$, is

$L = \{aaa, aab, aba, baa, abb, bab, bba, bbb\}$

Total 8 strings.

No. of Palindromes of **length 6**, defined over $\Sigma = \{ a, b \}$, is

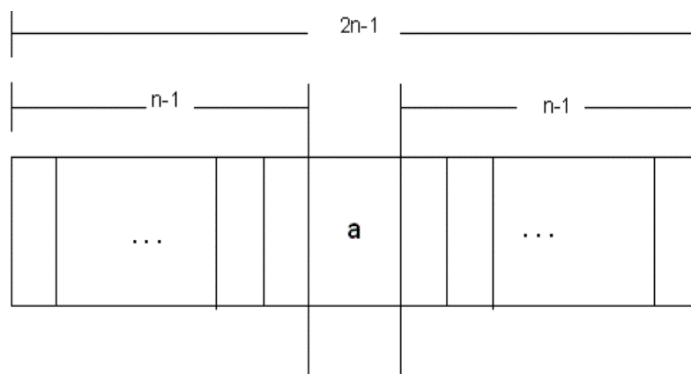
$PALINDROMES = \{aaaaaa, aabbaa, abaaba, baaaab, abbbba, babbab, bbaabb, bbbbbb\}$

Total 8 Palindromes.

Odd Length Case:

To calculate the total number of palindromes of length $(2n-1)$, need to calculate followings:

- Calculate the no. of palindromes of length $(2n-1)$, with “a” as middle letter.
- Calculate the no. of palindromes of length $(2n-1)$, with “b” as middle letter.
- Then add above two.



(if a is in middle) There are as many number of palindromes of length $(2n-1)$, as there are strings of length $(n-1)$.

No. of palindromes of length $2n-1$ = No. of strings of length $n-1$

For $n = 1$

No. of Palindromes of length 1 = No. of strings of length 0.

For $n=2$

No. of Palindromes of length 3 = No. of strings of length 1.

For $n=3$

No. of Palindromes of length 5 = No. of strings of length 2.

(if b is in middle) There are as many number of palindromes of length $(2n-1)$, as there are strings of length $(n-1)$.

No. of palindromes of length $2n-1$ = No. of strings of length $n-1$

For $n=1$

No. of Palindromes of length 1 = No. of strings of length 0.

For $n=2$

No. of Palindromes of length 3 = No. of strings of length 1.

For $n=3$

No. of Palindromes of length 5 = No. of strings of length 2.

For example:

Suppose $n = 4$,

$n-1 = 4-1 = 3$, means, all strings having **length = 3 or length = $n-1$** .

If $\Sigma = \{a, b\}$, then, total no. of strings having length **equal to 3 or length = $n-1$**
= $2^3 = 8$ strings

$L = \{aaa, aab, aba, baa, abb, bab, bba, bbb\}$

As $n = 4$,

$2n-1 = 8-1 = 7$, means, all palindromes having **length = 7 or length = $2n-1$** .

If $\Sigma = \{a, b\}$, then, total no. of palindromes having length **equal to 7 or length = $2n-1$**
= $2n-1 = 2^3 + 2^3 = 16$ palindromes.

a as middle

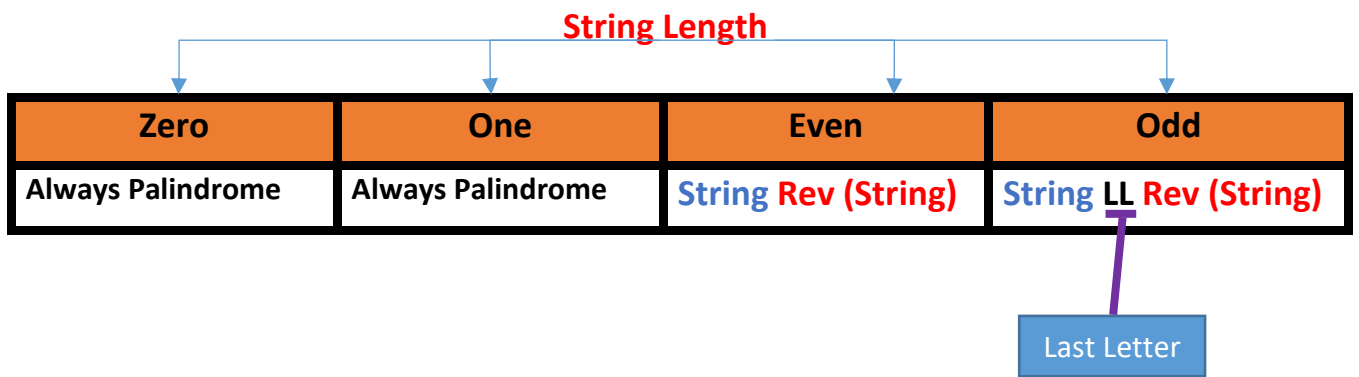
$L = \{aaaaaaa, aababaa, abaaaba, baaaaab, abbabba, bababab, bbbaabb, bbbabbb\}$

b as middle

$L = \{aaabaaa, aabbbaa, abababa, baabaab, abbbba, babbbab, bbababb, bbbbbb\}$

These are total, 16, means, 2^n , as $n = 4$.

How to make Palindrome Strings?



Question: If $\Sigma = \{aaB, c\}$, define a language of strings having length of strings 2, i.e., $|s| = 2$. How many number of strings in this language?
Also, define a language of palindromes having length 4. How many number of palindromes in this language?

Question: If $\Sigma = \{001, 10\}$, define a language of strings having length of strings 4, i.e., $|s| = 4$. How many number of strings in this language?
Also, define a language of palindromes having length 8. How many number of palindromes in this language?